

Computational Motor Control - 2026

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1 Introduction

Locomotion in vertebrates arises from the coordinated activity of neural circuits known as Central Pattern Generators (CPGs). In salamanders, these circuits are particularly well-studied due to the animal’s ability to transition seamlessly between aquatic swimming and terrestrial walking, making it an ideal model organism for understanding the neural control of locomotion.

This report presents the implementation and analysis of a CPG-based locomotion controller for a simulated salamander robot (Polymander), based on the coupled oscillator framework proposed by Ijspeert et al. [8], but further developed to include a higher number of degrees of freedom (DoF). Specifically, while in Project 1 the CPG controller was developed to uniquely produce Polymander’s swimming behavior, the goals of this second project were to implement the control of limbs to additionally reproduce the walking behavior of the salamander and to add sensory feedback in order to realize a close-loop locomotion controller.

The work is organized around four main sections. First, the limb-CPG network was implemented and validated in isolation, without physical simulation. Second, the model was embedded in a physics-based simulation environment to evaluate actual locomotion performance. Third, the coordination between limb and axial oscillators was systematically investigated by sweeping the limb-body phase offset and drive level and identifying parameter configurations that maximize speed and minimize energetic cost. Finally, sensory feedback was introduced to close the locomotion loop and allow the gait transition from swimming to walking, or vice versa.

2 Section 1: Limb-CPGs implementation

The goal of this first part was to implement the complete CPG network governing both the axial (spine) and limb joints of the Polymander robot, and to validate its dynamics without physics simulation. The implementation follows the oscillator model originally proposed by Ijspeert et al. [8], extended to account for the higher DoF robot used in this project. Each oscillator i is described by a coupled phase–amplitude system, where the phase dynamics are governed by:

$$\dot{\theta}_i = 2\pi f_i + \sum_j r_j w_{ij} \sin(\theta_j - \theta_i - \psi_{ij}) \quad (1)$$

where θ_i is the phase of oscillator i , f_i is the intrinsic frequency, w_{ij} the coupling weight between oscillators i and j , and ψ_{ij} the desired phase biases between them. The amplitude r_i converges exponentially to the nominal amplitude R_i , with rate a_i :

$$\dot{r}_i = a_i(R_i - r_i) \quad (2)$$

And the muscle output used for both axial and limb joints is:

$$M_i = r_i(1 + \cos(\theta_i)) \quad (3)$$

This output is always non-negative and oscillates between 0 and $2r_i$, making it directly suitable as a muscle activation signal for the Ekeberg antagonist muscle model used in this project [6].

The network of oscillators to drive Polymander’s locomotion was implemented to account for the limb-body coupling. Firstly, the connectivity matrices of coupling weights and phase biases were built as illustrated in Table 2. The axial CPG follows Table S1 of [8] exactly: ipsilateral nearest-neighbor coupling with weight $w = 10$ and phase bias $\pm 2\pi/8$ generates a caudally propagating traveling wave during swimming, while contralateral coupling ($w = 10$, $\psi = \pi$) enforces left–right anti-phase coordination, which is essential to reproduce alternating activation of antagonist muscle groups and thus generate balanced bilateral limb and trunk motions. The limb CPG departs from [8] but further accounts for the two-joint-per-limb architecture of Polymander: each limb has four oscillators, with antagonist pairs coupled in anti-phase ($w = 10$, $\psi = \pi$) and girdle–knee pairs coupled with a $-\pi/2$ phase bias and $w = 5$ to produce a quarter-cycle lag needed for a smooth circular leg trajectory. Inter-limb trot coordination (observed in real salamanders [2]) is achieved by coupling diagonal pairs in-phase and lateral pairs in anti-phase ($w = 10$, $\psi = -\pi$). Finally, the unidirectional limb-to-body coupling weight (stronger than the intrinsic body–body coupling, so that the limb CPG overrides the natural tendency of the axial chain to produce traveling waves, according to [8]) is set to $w = 150$, substantially larger than the value of 30 used in [8]. The larger value was required because Polymander has 8 actuated spinal joints instead of 6, making the collective traveling-wave tendency of the axial chain stronger and demanding greater limb influence to entrain all axial oscillators. The value was tuned empirically in the CPG simulations of Exercise 1. Only girdle oscillators (16, 20, 24, 28) project to the body, with the forelimb girdles reaching trunk segments and the hindlimb girdles reaching caudal segments, consistent with the anatomical locations of the cervical and thoracolumbar limb CPG centers shown

in Figure 1. The phase bias from limb to body oscillators is set via the parameter *limb_body_phase_offset* (default 0). As noted in the supplementary material of [8], the specific phase relationship that maximizes stride length is the one observed in real salamanders: the limb is maximally protracted when the corresponding trunk segment flexes toward the contralateral side; thus, this parameter is set to 0 accordingly.

The drive-dependent frequency and amplitude values were adapted from Ijspeert et al. [8]. They were implemented as piece-wise linear functions of the drive signal d , representing the descending tonic signal from the Mesencephalic Locomotor Region (MLR), a region in the brainstem that acts as the brain’s locomotor command center [14]. Different lower and upper thresholds were used for body and limb oscillators, summarized in Table 1. The lower saturation threshold for limb oscillators ($d = 3$ vs. $d = 5$ for body) is the central mechanism producing the gait transition from walking to swimming: once the drive exceeds 3, limb oscillators saturate ($R = 0$, $f = 0$), release their override on the body CPG, and allow it to resume the higher-frequency traveling wave pattern of swimming. Frequencies were converted to rad/s (multiplied by 2π) before being passed to the ODE (Ordinary Differential Equation) integrator, consistent with the phase variable θ_i being in radians. The nominal amplitudes of the axial oscillators were increased with respect to the original implementation of Ijspeert et al. [8]. In the original implementation, the CPG output $x_i = r_i(1 + \cos \theta_i)$ was used to compute a desired joint angle, which was then imposed by a PD (proportional-derivative) position controller, regardless of the mechanical state of the robot. In the present framework, the same CPG output instead sets the activation level of each Ekeberg antagonist muscle, and is then converted into muscle force and subsequently into joint torque. The actual joint angle that results depends on the torque balance between the two antagonist muscles, the passive joint stiffness, and the limb masses; it is not fixed by a controller. Larger nominal amplitudes are therefore required so that the muscles generate forces sufficient to produce joint displacements of comparable magnitude (the same value of R_i no longer corresponds to a proportional joint excursion), and consequently locomotor performance similar to the original implementation. Convergence rates are set to $a_i = 20.0s^{-1}$ for all oscillators, as suggested in supplementary material of [8].

The coupled ODE system (64 state variables: 32 phases and 32 amplitudes, consistent with SalamandraState) was integrated using the Dormand–Prince (dopri5) adaptive Runge–Kutta solver implemented in SciPy. The solver uses an adaptive internal time step controlled by absolute and relative error tolerances, while the system state was evaluated and logged at a fixed sampling interval of $\Delta t = 10ms$. Initial phases were set near zero to allow the network to converge toward its attractor from a synchronized initial condition.

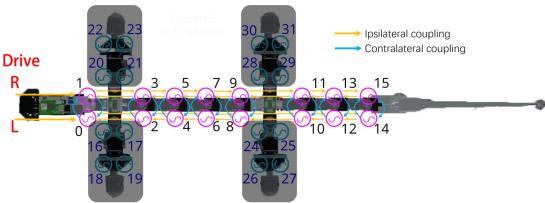


Figure 1: Polymander’s oscillators

	Body	Limb
No. oscillators	16	16
a_i [s^{-1}]	20.0	20.0
$[d_{low}, d_{high}]$	[1, 5]	[1, 3]
$[c_{v,1}, c_{v,0}]$ [Hz]	[0.2, 0.3]	[0.2, 0.0]
$[c_{R,1}, c_{R,0}]$ [rad]	[0.075, 0.8]	[0.15, 0.13]
v_{sat} [Hz]	0.0	0.0
R_{sat} [rad]	0.0	0.0

Table 1: Parameters used in the Polymander CPG model

In the first part of exercise 1, the Polymander’s locomotion was simulated with a fixed drive in the walking range ($d = 2.5$). Figure 2a shows the muscle signals for axial and limb joints. As expected, all axial joints connected to the left fore-limb (0,2,4,6,8) are in phase and in anti-phase with the joints connected to the corresponding hind-limb and contralateral fore-limb. Tracing a vertical line allows one to appreciate the typical standing wave expected from walking locomotion in the salamander (the trunk segments oscillate approximately in phase on each side while the tail oscillates in anti-phase, with nodes near the girdles). The diagonal pattern of trot gait is achieved with diagonal pairs of limbs in anti-phase (16 with 28, 20 with 24), lateral pairs in anti-phase, while knee joints are in a quarter-cycle lag with respect to their girdles. All antagonist pairs of muscles are correctly in anti-phase and the intrinsic frequency of all oscillators is constant over time due to the fixed drive.

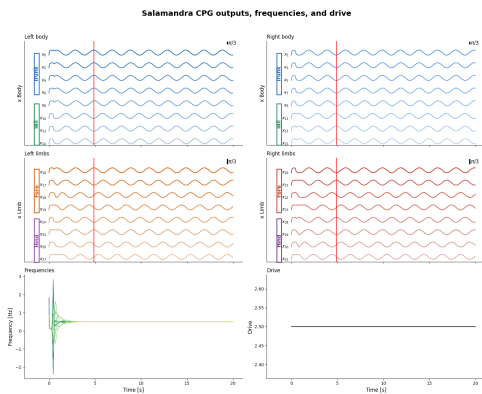
In the second part, a drive ramp linearly increasing from 0 to 6 (over 20 seconds) was used to drive the CPG network and observe the locomotion switch of Polymander, in accordance with Hypothesis 3 of [8]. As expected, in body joints, the signal is a standing wave for the walking regime ($1 \leq d \leq 3$), and it switches to a traveling wave for swimming values of the drive ($3 \leq d \leq 5$) (Fig. 2b). Accordingly, limb joints show oscillations only in the walking range of drives (since they saturate around $d = 3$), and the signal becomes flat when the robot starts to swim. From Fig. 2b one can also appreciate the increasing frequency of oscillations: with a linearly growing drive, the speed of locomotion (both walking and swimming) increases due to a higher descending signal from MLR,

as can also be observed by the plot of instantaneous frequencies (linearly increasing, with a flat signal of limb oscillators during swimming). A brief transient de-synchronization is visible as a spike in instantaneous frequencies before re-entrainment at the higher body frequency, reproducing the step increase of frequency characteristic of the walk-to-swim transition [4]. No oscillations are present in either body or limb for saturation values of the drive. These plots reliably match those showed in the original work of Ijspeert et al. [8]. Additional figures showing the convergence of specific pairs of oscillators' phase biases are included in the supplementary material to further prove the alignment between results and expectations.

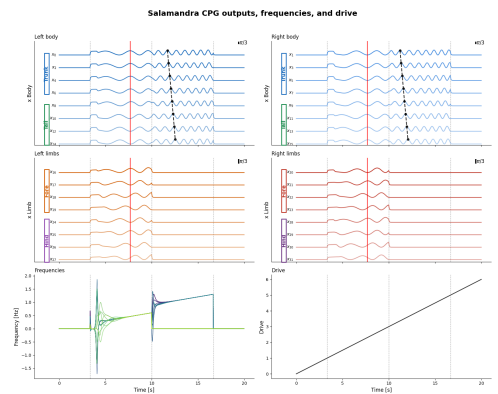
Table 2: CPG coupling weights w_{ij} and phase biases ψ_{ij} for all connection types in the Polymander network. Oscillator indices follow the numbering in Figure 1.

Connection type	Oscillator pairs ($i \rightarrow j$)	w_{ij}	ψ_{ij}
<i>Axial body CPG</i>			
Ipsilateral, descending (L)	$2k \rightarrow 2k+2, k = 0, \dots, 6$	10.0	$-2\pi/8$
Ipsilateral, ascending (L)	$2k+2 \rightarrow 2k, k = 0, \dots, 6$	10.0	$+2\pi/8$
Ipsilateral, descending (R)	$2k+1 \rightarrow 2k+3, k = 0, \dots, 6$	10.0	$-2\pi/8$
Ipsilateral, ascending (R)	$2k+3 \rightarrow 2k+1, k = 0, \dots, 6$	10.0	$+2\pi/8$
Contralateral (same segment)	$2k \leftrightarrow 2k+1, k = 0, \dots, 7$	10.0	π
<i>Limb CPG - intra-limb</i> (base index $b \in \{16, 20, 24, 28\}$)			
Girdle antagonists	$b \leftrightarrow b+1$	10.0	π
Knee antagonists	$b+2 \leftrightarrow b+3$	10.0	π
Girdle-knee (extensor side)	$b \leftrightarrow b+2$	5.0	$-\pi/2$
Girdle-knee (flexor side)	$b+1 \leftrightarrow b+3$	5.0	$-\pi/2$
<i>Limb CPG - inter-limb</i> (girdle oscillators only)			
Fore/hind pairs (trot, anti-phase)	$16 \leftrightarrow 20, 24 \leftrightarrow 28$	10.0	$-\pi$
Lateral pairs (trot, anti-phase)	$16 \leftrightarrow 24, 20 \leftrightarrow 28$	10.0	$-\pi$
<i>Limb-to-body</i> (unidirectional, girdle $\rightarrow$ body)			
LF girdle (16) $\rightarrow$ trunk L	$16 \rightarrow \{0, 2, 4, 6, 8\}$	150.0	0
RF girdle (20) $\rightarrow$ trunk R	$20 \rightarrow \{1, 3, 5, 7, 9\}$	150.0	0
LH girdle (24) $\rightarrow$ tail L	$24 \rightarrow \{10, 12, 14\}$	150.0	0
RH girdle (28) $\rightarrow$ tail R	$28 \rightarrow \{11, 13, 15\}$	150.0	0

Notes. $\leftrightarrow$ denotes bidirectional coupling (both directions carry the same w_{ij} and ψ_{ij}). Abbreviations: L = left, R = right, LF = left forelimb, RF = right forelimb, LH = left hindlimb, RH = right hindlimb.



(a) CPG output for a fixed drive ($d = 2.5$)



(b) CPG output for a ramp drive ($0 \leq d \leq 6$)

Figure 2: Limb-CPG outputs

3 Section 2: Physics simulation

In this second section of the project the CPG network was connected to robot physics model, to demonstrate basic locomotion.

3.1 Physics integration

A fixed drive of $d=2.5$ was applied, placing the system within the walking regime. At this drive level, the body oscillates with frequency $f_{\text{body}} = 0.8$ Hz and nominal amplitude $R_{\text{body}} \approx 0.89$, while the limb oscillators run at $f_{\text{limb}} = 0.5$ Hz with $R_{\text{limb}} \approx 0.51$. The inter-limb coupling, with diagonal pairs in-phase and ipsilateral pairs in anti-phase, produces a stable trotting gait with coordinated body undulation. The robot moves forward, validating that the full 32-oscillator network drives physically plausible locomotion, showed in the relative video (Python\videos\ex2a_walk\video_walking.mp4).

3.2 Drive ramp experiment

A linear drive ramp was implemented, starting from null value to saturation values over 40 seconds. The experiment was performed independently on flat ground and in water. At each time step, the drive modulates oscillator frequencies and nominal amplitudes according to the piecewise-linear relations from [8]. Outside each range, the corresponding frequency or nominal amplitude is set to zero, silencing the relevant oscillator group.

As the drive increases, three distinct behavioral regimes can be observed. With drives below the lower bound of 1 the salamander seems quiescent: all oscillators remain sub-threshold, neither limb nor motion is produced, and the robot remains stationary. The second behavior is walking trot, (drive $1 \leq d < 3$), in which both axial and limb oscillators activate simultaneously. The limb CPG produces a trot pattern by having diagonal limb pairs in phase, while ipsilateral pairs oscillate in anti-phase. On land, this translates into forward propulsion with stable four-limb ground contact. In the trot of real salamanders the ventral aspect of the trunk is always cleared of the ground and doesn't enter contact with it [9], which is coherent with the behavior observed in the Polymander simulations of walking. When in water, applying this same movement pattern doesn't allow the salamander to move efficiently, compared to the terrestrial environment, blocking the locomotion of the animal.

The next observed regime is swimming (drive $3 \leq d \leq 5$). Limbs oscillators are silent, while the body oscillators continue to increase both in frequency and amplitude, producing the full traveling wave that characterizes salamander locomotion during swimming. In particular, other than silencing the limb by setting frequencies and nominal amplitudes to zero, the controller explicitly overrides the joints offset target for all four limbs at $-\pi/2$ in the state array, this drives each limb joint toward the folded-back position, actively folding the limbs backwards during swimming (implemented in `salamandra_simulation/controller.py`). This ensures the limbs are aerodynamically retracted and held steady, reducing drag during anguilliform body undulation while swimming, which is done naturally in real salamanders. On land, the absence of limb support prevents the robot from effective forward locomotion, allowing only an undulation against the ground. In water, on the other hand, traveling waves represent the dominant and most efficient propulsion mode. The last behavioral regime is represented by saturation, when the drive exceeds values of 5, the body oscillators become silenced again, and all motion ceases. The gait transition at $d \simeq 3$ matches the exact threshold indicated in [8], and is consistent with the previously analyzed dynamics, showing that while transitioning from walking to swimming, the movements of limb oscillations vanish while axial oscillations increase in amplitude and frequency.

Experimenting with each behavioral regime both on land and in water allows for the investigation of the usefulness and effectiveness of each gait in different environments. On land, trotting represents the productive locomotor mode, with four-limb ground contact supporting the posture and pushing forward. Moreover, the S-shaped standing wave body undulation that accompanies walking is not merely passive: it actively increases stride length by flexing the trunk in coordination with limb protraction. In the absence of leg contact, the body undulation of the traveling wave is insufficient to produce any forward movement on land, while in water the opposite occurs. While swimming, the limbs generate a drag effect; thus, relying solely on the traveling wave seems to be the optimal strategy. This is consistent with the biomechanics of real salamanders.

Real salamanders display a richer variety of dynamics that the current model is not able to replicate. In the biological literature other gaits are documented, such as lateral-sequence walk, crawl or belly-drag, and a gradual gait transition involving limb retraction [1, 7]. At slow speeds salamanders use sequential walking gait with one lifted foot at a time [3]. This footfall pattern cannot be reproduced by the current inter-limb coupling. Furthermore, sometimes real salamanders use crawling as a secondary movement on land, which is very similar to swimming movements performed on the ground with the limbs folded against the body. Our model has no

feedback from the terrain to induce this behavior.

The primary missing elements in the current robot implementation are sensory feedback, with proprioceptive signals encoding the limb load and ground contact that would allow gait transition; richer inter-limb connectivity, enabling lateral-sequence or other patterns; supraspinal modulation beyond the MLR, given that descending pathways from cerebellum and basal ganglia shape context-dependent gait selection better than a single global drive.

The appropriate gait is selected primarily by MLR, which provides tonic excitatory drive to the spinal CPG, graded in amplitude. This single parameter is sufficient to select between the two principal locomotor modes (low values for walking and high values for anguilliform swimming). The brain can therefore exploit this mechanism for environment-appropriate gait selection. When entering water, a salamander increases MLR activity, shifting d above 3 and triggering the transition to swimming. When returning to land, the drive decreases, reactivating the limb CPGs and restoring walking. Recruiting more specialized gait variants, such as a lateral-sequence walk at very low speeds, would likely require either finer grading of the same drive signal or additional descending pathways targeting specific spinal interneuron populations, rather than uniformly exciting the entire CPG network.

4 Section 3: Axial-limb coordination and optimization

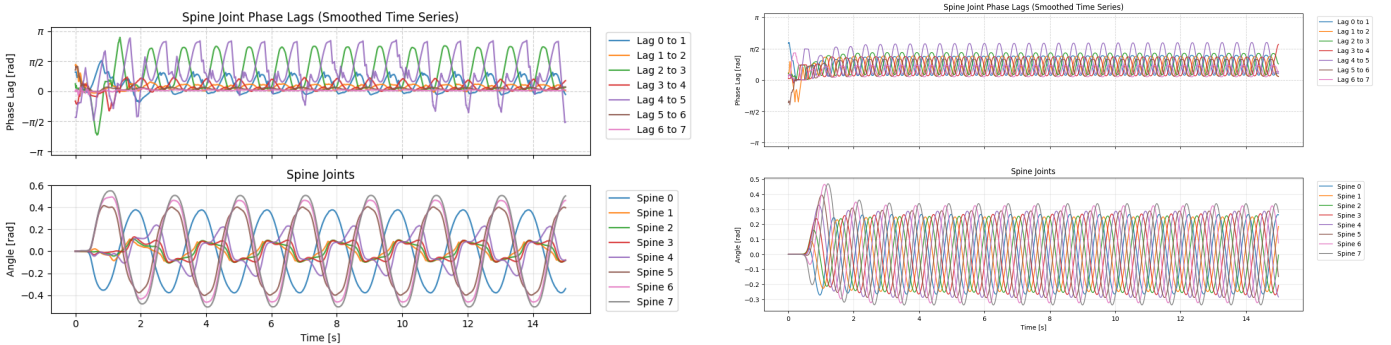
4.1 Spine phase analysis and role of couplings

During the walking gait, the robot employs the imposed diagonal trot and its spine segments move synchronously in a fixed "S" shape. From Figure 3a we can observe that the spinal segments follow a rhythmic pattern: the upper part (joints 1 to 3) and the lower part (joints 5 to 7) of the spine are synchronized. This is further seen in the phase lags of adjacent segments (Figure 3a) that are mostly at zero, showing the effect of a standing wave. This behavior is not present between joints 4 and 5 due to the segments being coupled with different sets of limbs, thus the effect was expected. The behavior is also not present between joints 0 and 1, and joints 2 and 3. The first scenario could be explained by the direct connection between the spine segment of the fore limbs and the head, thus acting like the unanchored extremities of the S-wave; while the second scenario is probably the result of the mathematical approximation: the phase lag of the spine has been calculated from kinematic data in the joint position and joint velocity plane $\langle \theta, \dot{\theta} \rangle$, where the phase is

$$\phi = \arctan \left(\frac{\dot{\theta}}{\theta} \right). \quad (4)$$

The velocity of the upper joints are not matching, causing spikes in the phase lag plots. However in the joint angle plot we can qualitatively see that they are synchronized in a standing wave.

During swimming gait, the robot deploys a travelling wave with a gradient of amplitudes starting from the head to the tail of the robot, where the joints of the spine keep a rhythmic evolution of phase lag with each other (Fig. 3b). The overall behavior of the robot matches real salamander data. Salamanders employ diagonal walking trot and the spine stretches into a fixed "S" shape to maximize stride length, while during swimming gait salamanders employ anguilliform swimming similarly to the lamprey [8]. This duality in locomotion control is only possible due to the presence of a strong, unilateral coupling between limb and body oscillators in the CPG.



(a) Time evolution of spine joint phase lags (above) and spine joint angles (below) during walking gait. (b) Time evolution of spine joint phase lags (above) and spine joint angles (below) during swimming gait.

Figure 3: Spine joints phase lags

To validate such claim, a simulation disabling the unilateral limb to body coupling was performed and saved (Python\videos\ex3.2_walk\video_walking.mp4). The robot fails to deploy a stable walking gait, showing no synchronization between limbs and trunk. In Figure 4 the joint angles appear to be a hybrid between the standing wave (walking gait) and a travelling wave (swimming gait); the phase lags plot shows no rhythmic cycle. This highlights the importance of a strong, unilateral limb to body neural coupling for locomotion control.

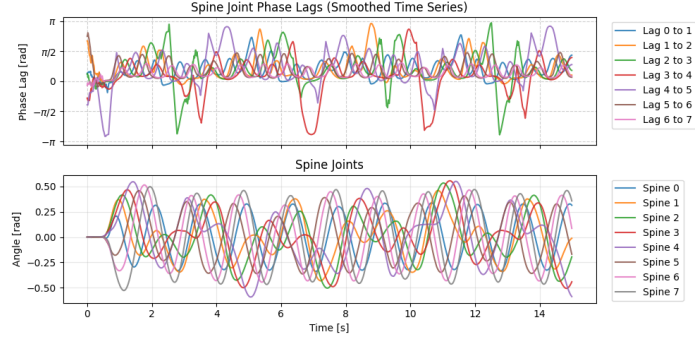


Figure 4: Time evolution of decoupled salamander between limbs and trunk. Spine joint phase lags (above) and spine joint angles (below) during walking gait.

To analyze the stability of the salamander, the following metrics were employed: forward speed, cost of transport (CoT), lateral path deviation, phase locking error and heading circular variance. More details regarding metrics computation is included in the supplementary material.

In Table 3 we can observe that the implementation without the unilateral coupling between limb and body performs worse on every single metric of interest. In particular, the locomotion generated by the decoupled CPG leads to a much more unstable trajectory where most of the momentum is spread across different directions; this is the main cause of a lower forward speed, higher lateral path deviation, and higher heading circular variance compared to the coupled implementation. The decoupled implementation also has higher phase-locking error, caused by the travelling wave component of the axial body that makes difficult to maintain the nominal phase offset initialized in the CPG.

	Metric	Coupled	Decoupled
	Forward speed [m/s]	0.2526	0.0699
	CoT [J/(N m)]	0.0002	0.0004
	Lateral path deviation [rad]	0.0455	0.1566
	Phase-locking error (RMS) [rad]	0.9023	1.8789
	Phase-locking error (MAE) [rad]	1.0178	2.1133
	Heading circular variance [rad]	0.1207	0.7005

Table 3: Summary of metric performance of salamander robot in walking gait on land: presence of unilateral coupling between limb and body.

4.2 Limb-body phase offsets

To analyze the impact of different limb-body offset with different values of drive in walking regime, a 2D grid search was made: we analyzed offsets ranging from $[-180, 180]$ degrees with a resolution of 18 degrees, and a range of $[1.00, 3.00]$ drive values with a resolution of 0.2. The metrics used are forward speed and CoT, defining parameters as optimal if they maximize forward speed (since CoT had overall low values).

In Figure 5a we can observe that forward speed is mainly dependent on drive with direct proportionality and it is higher at offsets ranging from 40 to 100 degrees. However, at negative offsets ranging from -120 to -50 degrees, forward speed is heavily degraded regardless of drive value. CoT follows a similar trend, however it minimizes even at lower levels of drive too. The optimal parameters chosen are $(66^\circ, 3.0)$ which maximize forward speed at 0.3553 m/s and have a CoT of <0.0000 .

To validate the finding, a simulation with the aforementioned parameters was executed and saved at the path (Python\videos\3.3_walk\video_walking.mp4), and also another simulation but in anti-phase was done and saved (Python\videos\3.3_walk\video_walking_antiphase.mp4). The optimal parameters show the robot deploying a correct walking gait, where the spine moves only after the limbs are anchored onto the ground, creating forward momentum and propelling the body forward. Conversely, the anti-phase simulation shows a

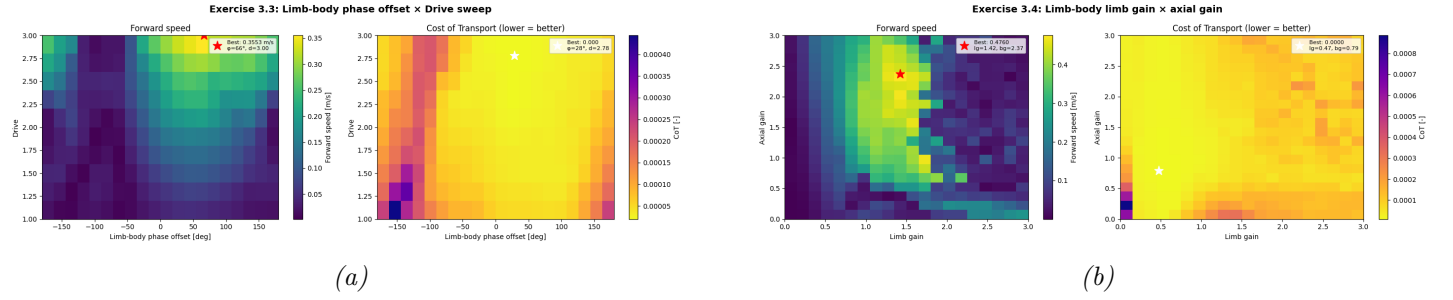


Figure 5: 2D parameter sweeps evaluating forward speed and CoT. (a) Sweep varying drive values and limb-body offset. (b) Sweep varying amplitude gains of body and limb oscillators.

total depletion of locomotion control, where the robot fails to deploy an effective walking gait and travelling in random directions in the virtual environment. This is caused mainly by the trunk swaying the opposite direction of the anchored limbs, canceling any forward momentum generated and causing the robot to bounce and veer. The optimal locomotion found is compatible with biological findings: during terrestrial trot, the swing of the limb anticipates the lateral bending of the body, causing the spine to finish bending when the foot is already planted onto the ground. This artificially extends the stride length and is stable [8].

4.3 Amplitude gains

Once the optimal drive and limb to body phase offset were found, we proceeded in a similar fashion to investigate the impact of amplitude gain in body and limb oscillators: the amplitudes were searched in a grid ranging [0, 3] with a resolution of 0.15.

In Figure 5b we observe that forward speed is mainly dependent from limb amplitude gain: forward speed increases proportionally to limb gain, with axial gain not influencing the metric especially when oscillators are still well bounded within their nominal amplitudes. From limb gain greater than 0.65, higher axial gain also leads to higher forward speed. This behavior drastically changes at limb gains of 1.5 and higher: forward speed is degraded representing failure in locomotion, caused by applying excessive torque from the limbs lifting the salamander too much onto the air. CoT appears to be low uniformly across oscillator limbs, except for extremely low values of axial gains paired with low values of limb amplitude gains, showing insufficient torque to move the robot on ground.

The optimal parameters are $limb\ gain = 1.42$ and $axial\ gain = 2.37$, causing the robot to sustain stable walking gait at 0.4760 m/s. However, in the video (Python\videos\3.4_walk_video_walking.mp4) we observe that the CoM of the salamander also shifts in the z-plane: the large torque amplified by the limb gain causes the body to lift itself from ground. This behaviour is not found in biological salamanders and it highlights a limitation of the physics simulation: the robot is approximated by rigid-body actuators, lacking the complex anatomical components of the musculoskeletal system such as joint capsules, ligaments and tendons [9]. This level of approximation allows the actuators to apply unnatural levels of torques, so by prioritizing metrics such as forward speed, we might loose natural walking gaits.

5 Section 4: Gait transition

In this section we explored the role of sensory feedback as a mechanism of gait switching. The MuJoCo framework is capable of retrieving robot-environment collision data: we used the method `data.sensors.contacts.totals` to retrieve the magnitude of contact forces categorizing them based on which anatomical region they involved (body, proximal limb, and feet). We imposed the gait switching mechanism as a dynamical drive update based on the contact forces of the feet: if there is any contact force applied on the feet, we assume that the robot has enough space on land to deploy walking gait by updating its neural drive; otherwise, the robot deploys swimming gait. This feedback loop update is executed every 50 iterations to avoid spurious switching between gaits, translating into a 250 ms loop with a timestep of 0.005 s.

Figure 6a shows the time evolution of the joint angles, while Figure 6c shows the contact forces during the transition between swimming and walking gait (Python\videos\ex4_swim2walk_video_swim2walk.mp4). The robot successfully deploys initial swimming gait and switches to walking gait when it detects contact force at their feet at around 29 seconds. In particular, the majority of the contact force is concentrated at the feet of the forelimbs, with body joints and proximal limb joints having near zero contact forces during walking gait. This finding is compatible with biological studies of the salamander, where the forelimbs were found to be responsible of the

majority of the vertical deceleration, in the form of weight bearing, and longitudinal braking; while the tarsus never touches the ground during stance. This makes the salamander stop the forward and downwards momentum resulting in instantaneous big spikes collision forces [9].

Although walking gait initiates at around 29 seconds, the forelimbs began to sense contact forces at around 26 seconds. Actually, there are some small activations of the girdle joints at around 26 seconds, but the spine still shows travelling wave component of swimming gait. This is a potential limitation of our feedback mechanism implementation, where instead of local CPG inhibition/activation based on sensory feedback, we directly switch the global neural drive. This is much more evident in Figure 6b that describes the transition from walking to swimming (Python\videos\ex4_walk2swim\video_walk2swim.mp4): at around 22 seconds, when the salamander reaches the shallow waters, the forelimbs gradually stop receiving contact forces and are unable to propel the body forward. In this time frame, the robot still maintains walking gait since hindlimbs still receive feedback information, however they are not as capable of generating forward propulsion. The robot deploys swimming gait only when all contact feedback is ceased, well beyond the shallow waters of the arena. This highlights one of the limitations of the implementation, suggesting the need to design at least a threshold activation function.

Biological salamanders transition gaits through a distributed integration of local sensory feedback and descending neural control. Terrestrial walking relies on proprioceptors across all four limbs to detect mechanical loading, while swimming utilizes the lateral line system to sense hydrodynamic forces. Although the brainstem regulates global neural drive, gait switching is heavily mediated by local reflexes; mechanically unloading the limbs directly inhibits the limb CPGs, tucking the legs and allowing the axial CPGs to take over [8][12]. In contrast, our implementation utilizes a binary global switch based solely on feet collision forces. While it works for the simple provided amphibious arena, our implementation probably fails with more complex terrain geometry, such as uneven terrain within land and water.

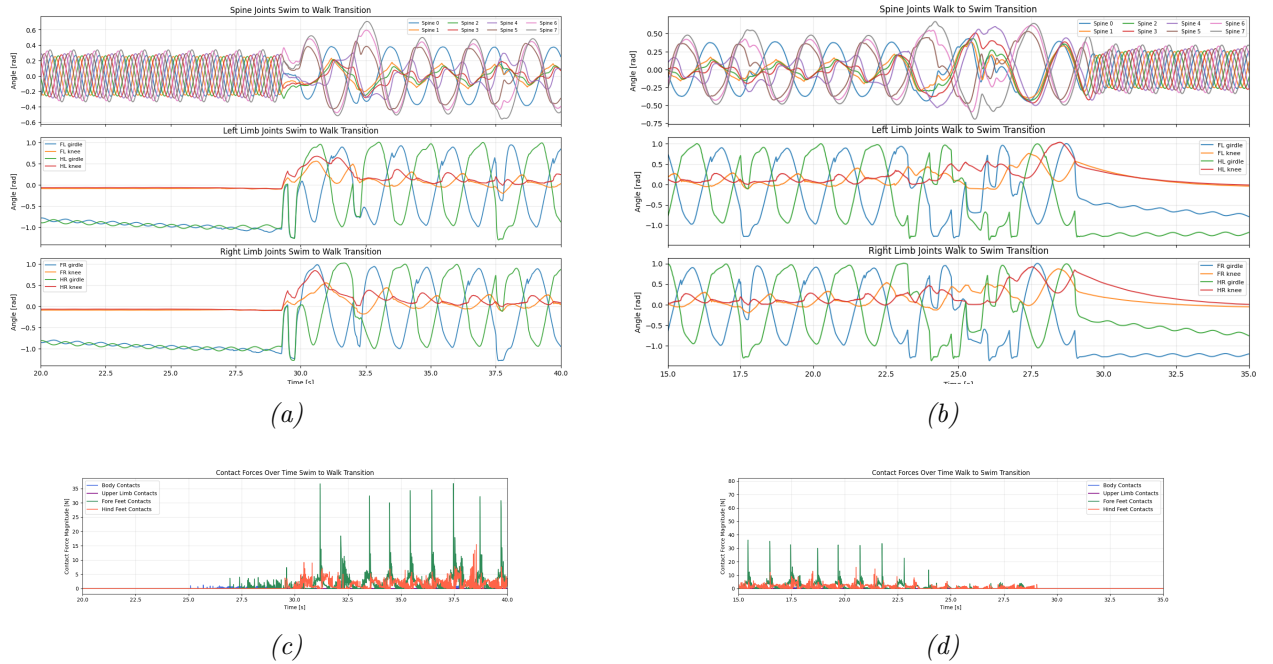


Figure 6: Time evolution of joint angles and contact forces of the robot during gait transitions. The left column (a, c) displays the swim-to-walk transition, while the right column (b, d) displays the walk-to-swim transition.

6 Section 5: Future developments

Despite the presence of several studies on salamander locomotion, quantitative kinematic data remain limited, particularly for terrestrial gaits and limb-body coordination [9]. Multiple interacting factors influence salamander motor control, including specific species, environmental conditions, and underlying mechanics. Because these variables are difficult to manipulate independently in vivo, a computational neuromechanical framework such as the one developed in this project provides a controlled environment in which individual mechanisms can be isolated and systematically tested. The developed framework could be used to address three main research questions:

1. What functional role does the rostro-caudal traveling body wave play during terrestrial walking in salamanders?

2. Do forelimbs and hindlimbs contribute differently to propulsion and stability during walking?
3. Does spatially varying curvature amplitude along the body improve terrestrial locomotion performance compared to uniform curvature?

The first question concerns the relevance of axial traveling waves during walking. Although traveling waves are classically associated with swimming, Ryczko et al. [15] demonstrated that axial CPGs can continue to exhibit traveling-wave activity even when limb CPGs are active. This suggests that axial rhythmicity is not simply suppressed during terrestrial locomotion, but may retain a functional role. Based on this evidence, we hypothesize that a rostral-caudal traveling wave during walking enhances dynamic stability and reduces limb mechanical effort compared to a rigid trunk or standing-wave pattern [9]. To test this hypothesis, the axial phase lag between trunk segments could be systematically varied while keeping limb control constant (e.g., no phase lag, moderate phase lag, exaggerated lag). Performance metrics such as cost of transport, lateral path deviation, phase-locking error, and heading circular variance would be quantified. If a specific phase lag minimizes energetic cost or improves stability, this would support the idea that axial undulations in walking are functionally advantageous. Biologically, this would clarify whether the persistence of traveling waves reflects an adaptive coordination strategy or simply intrinsic spinal dynamics.

The second question addresses potential functional specialization between forelimbs and hindlimbs. While experimental data on salamander walking demonstrate coordinated axial-limb motion, quantitative comparisons of forelimb versus hindlimb contributions to propulsion and stability are lacking [9]. In many sprawling tetrapods, hindlimbs contribute more to propulsion, whereas forelimbs contribute to braking and postural stabilization [13]. We therefore hypothesize that hindlimbs primarily generate propulsive impulse, while forelimbs contribute more strongly to stability control and braking. Computationally, this could be tested by selectively reducing torques or modifying phase relationships in either forelimbs or hindlimbs, and evaluating the resulting changes in forward speed and stability. If reducing hindlimb torque significantly decreases speed while reducing forelimb torque increases instability, this would support functional specialization. Such findings would provide quantitative insight into limb differentiation in salamander locomotion.

The third question concerns the spatial distribution of axial curvature along the trunk. Experimental observations indicate that curvature amplitude during walking is not necessarily uniform along the body axis [5, 9]. However, the mechanical implications of this non-uniformity remain unexplored. We hypothesize that posterior-amplified curvature enhances coordination and improves propulsion efficiency compared to uniform curvature. To test this, different curvature profiles (uniform amplitude versus posterior-amplified bending) could be implemented while keeping limb control fixed. Metrics such as cost of transport and forward speed would be compared. If posterior bending reduces overall energetic cost or increases forward speed, this would suggest that spatial modulation of trunk curvature is a mechanically optimized feature. Biologically, this would improve our understanding of axial-limb synergy and may help explain salamander locomotion kinematics.

Overall, these investigations highlight how a computational model can serve as a hypothesis-testing platform to disentangle neuromechanical interactions that are experimentally difficult to isolate. By systematically manipulating axial phase lags, limb contributions, and curvature distributions, the framework can generate quantitative predictions to guide future experimental and robotic studies of salamander locomotion.

7 Conclusion

This project implemented and analyzed a neuromechanical model of salamander locomotion based on coupled CPGs, extended to a high-dimensional amphibious robot. The results show that a simple oscillator-based architecture can reproduce key locomotor behaviors, including terrestrial trotting, swimming with traveling body waves, and smooth gait transitions driven by a single descending drive signal.

Across simulations, the system exhibited coherent limb-body coordination, with biologically plausible phase relationships enabling switching between locomotor modes. The physics simulation showed how neural activity maps to mechanical output, with limb-trunk coupling, phase offsets, and amplitude modulation.

Parameter sweeps and ablation studies showed sensitivity to limb-body phase relationships and actuation gains, while revealing limitations such as missing musculoskeletal constraints and sensory feedback, suggesting feedback-driven control and more flexible gait selection as improvements.

Overall, the results support CPG-based control for amphibious locomotion and provide a framework for studying vertebrate motor control, bridging simplified neural architectures with biological complexity.

8 Supplementary material

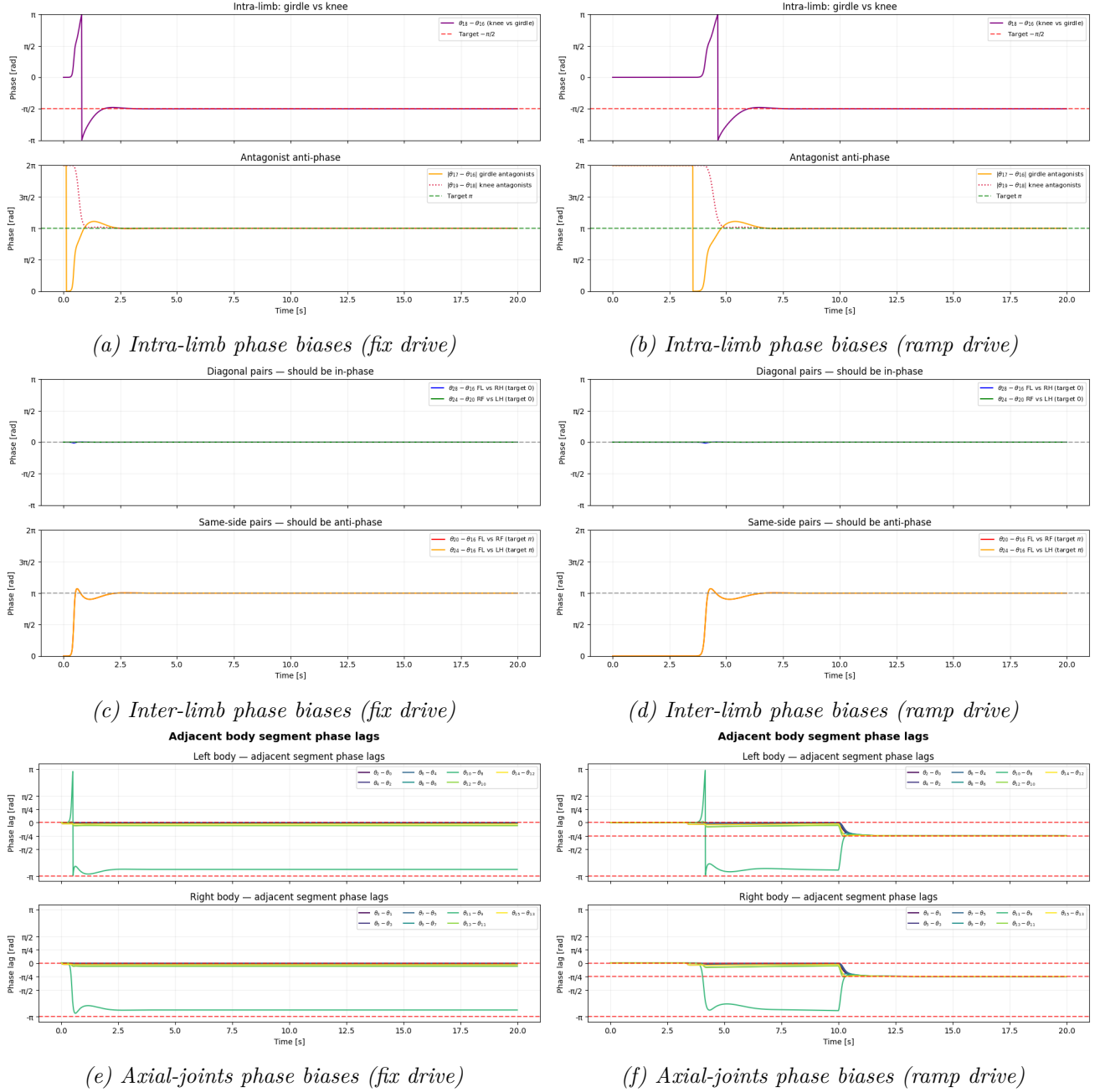


Figure 7: CPG phase biases for fixed drive ($d = 2.5$, left column) and linearly increasing drive ($d : 0 \rightarrow 6$ over 20 s, right column). (a,b) Intra-limb phase relationships: the girdle-knee phase difference $\theta_{18} - \theta_{16}$ converges to the target $-\pi/2$ (top), and antagonist pairs converge to anti-phase π (bottom), confirming correct intra-limb coordination. (c,d) Inter-limb trot diagnostics: diagonal pairs (LF-RH and RF-LH) lock to in-phase (top), while same-side pairs (LF-RF and LF-LH) lock to anti-phase π (bottom), validating the trot coupling architecture. (e,f) Adjacent axial segment phase lags: under fixed drive all segments maintain a constant lag near 0 per segment (standing wave during walking), while under the drive ramp the phase lag transitions from 0 (standing wave, walking regime) to $-\pi/4$ per segment (traveling wave, swimming regime) once the limb oscillators saturate at $d \approx 3$. Trunk and tail oscillators are nearly in anti-phase (phase-bias close to $-\pi$ during the walking regime to produce a S-wave).

Forward speed was calculated as the average speed of the center of mass (CoM) during the time window of interest along the y axis. CoT was calculated as $CoT = \frac{(\tau \dot{v})}{mgd}$, where $(\tau \dot{v})$ is the energy spent to generate the movement as a function of joint torque and joint velocity, mg the weight of the salamander, and d is the CoM forward distance; in ideal locomotion the animal minimizes this metric. Lateral path deviation was calculated as the root mean square (RMS) perpendicular distance from the ideal straight-line trajectory connecting starting and ending position during the time window of interest; ideal locomotion yields an overall straight trajectory, with higher values indicating wandering and reduced locomotor stability [8]. Phase-locking error represents the ability of the

robot to keep the desired limb-body phase offset defined in CPG initialization and both RMS and mean absolute error have been calculated; the phases are extracted with the already described Equation 4. In ideal locomotion, phase-locking error is minimized and it is commonly used in CPG theory [11]. *Heading circular variance* measures the grade of head fluctuations of the robot, with higher values representing directional instability. The angle of the head direction (yaw) is extracted using $\theta_{\text{yaw}} = \arctan(v_x/v_y)$. Variance computation cannot be applied to angles since everything wraps between $[-\pi, \pi]$, so we implemented circular statistics [10]: we first projected the yaw angles onto the unit circle, averaged them over time, and computed the resultant

$$\begin{aligned} C &= \frac{1}{T} \cos(\theta_{\text{yaw}}) \\ S &= \frac{1}{T} \sin(\theta_{\text{yaw}}) \\ R &= \sqrt{C^2 + S^2} \\ \text{Var} &= 1 - R. \end{aligned}$$

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